

FZ9

$$f(x) = \frac{\cos x - 2x}{\sin x}, \quad x \in (0, \pi)$$

$$(\sin x)' = \cos x \quad (\cos x)' = -\sin x$$

① wyznaczamy wzór na $f'(x)$, korzystając z pochodnej ilorazu

$$\left[\frac{f(x)}{g(x)} \right]' = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{[g(x)]^2}$$

$$\begin{aligned} f'(x) &= \frac{(-\sin x - 2)\sin x - \cos x(\cos x - 2x)}{\sin^2 x} = \frac{-\sin^2 x - 2\sin x - \cos^2 x + 2x\cos x}{\sin^2 x} = \\ &= \frac{-(\sin^2 x + \cos^2 x) - 2\sin x + 2x\cos x}{\sin^2 x} = \frac{2x\cos x - 2\sin x - 1}{\sin^2 x} \end{aligned}$$